

5

Strategies and Challenges in Machine learning in healthcare

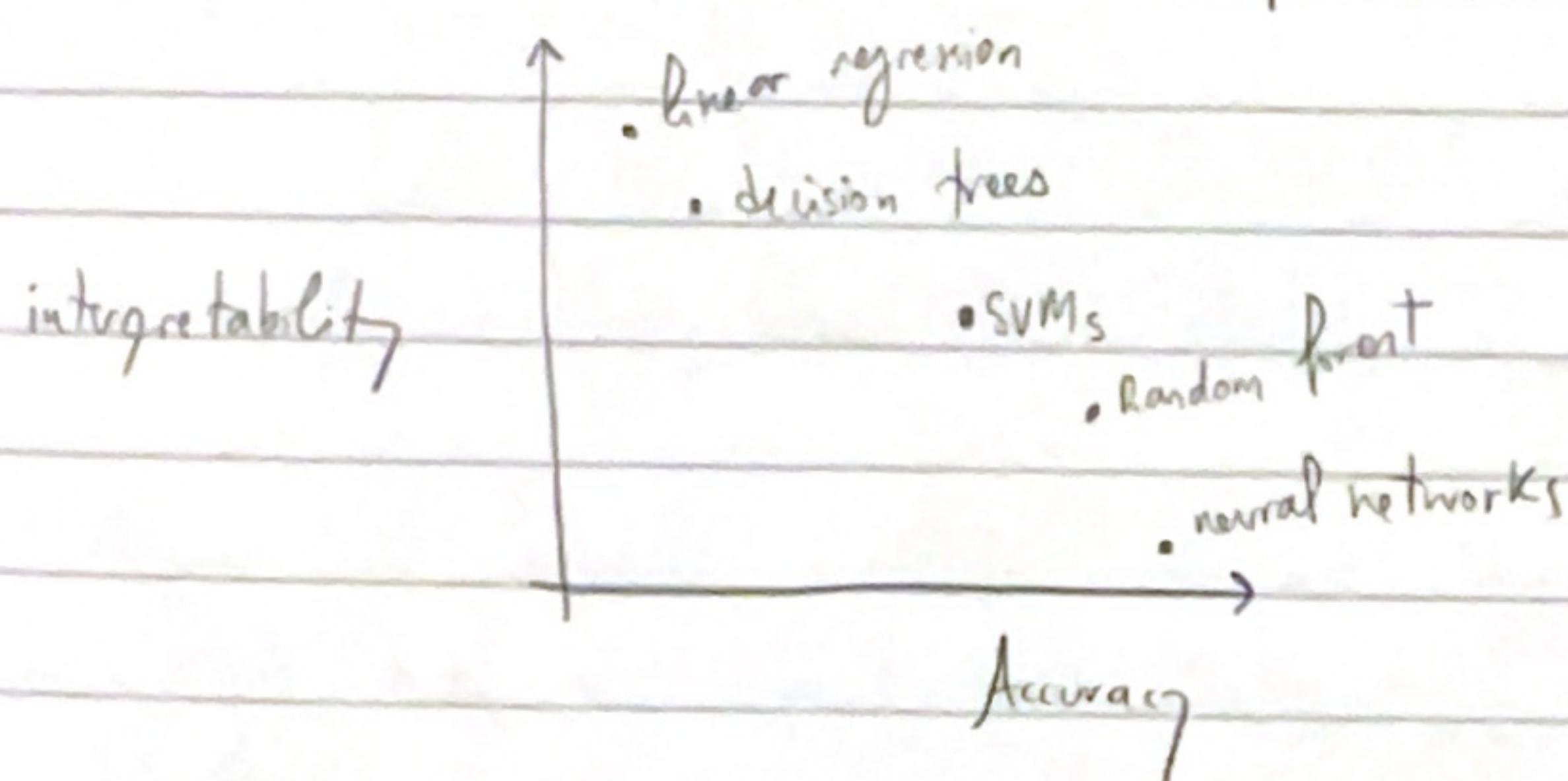
Unforeseen variables:

- patterns can be — correlations — (— common response variables)
 - confounding factors
 - lurking variables

→ Reconfigure model

Context in Clinical ML

→ T between black box and interpretable algorithms



Black boxes → we don't understand the algorithm

→ we observe Inputs / Outputs → but not internal workings

⇒ efforts to bring more interpretability

→ leveraging MOT to review false +
false -

→ test the model to external datasets → try to gain insight
into causal vs

correlative features

⇒ SALIENCY → part that matters the most to the model to make
prediction

Class activation maps → based on analysing the neurons in the
(CAMs) final layer of some types of neural

networks to compute how much the
neurons that are important for any particular class are firing at every spatial location

1° primary
scores

2° importance

→ produce bootstraps

→ Computing change in prediction score from changing something

→ Transparent → Can lead to overconfidence

→ explainable

→ inspectable

Intrinsic interpretability → simple, self explanatory models

Post-hoc " → used to understand decisions by complex models that don't have prescriptive declarative knowledge representations or features

Medical Data Challenges in ML

data types:

- EHR
- structured / unstructured
- imaging
- genomics
- peripheral sources

→ small representative sample (of the type) → understand how it's generated $\xrightarrow[\text{how}]{\text{consider}}$ it fits into ML workflow

get a sense of
how much data $\xleftarrow[\text{is needed}]{\text{generate}}$ Metrics

Sample
Preliminary analysis
Discussion } Should be done before investing many resources

→ data used to train model → Should be available in real world

what preprocessing
feature engineering of data will be needed

→ Data in medicine can be

- heterogeneous
- incomplete
- duplicated

① Missing data

→ class imbalance → in amount of labels

↓
accuracy paradox problem → ↑ accuracy → doesn't reflect "real" accuracy

↓
they just choose the
label with ↑ amount
(over represented class)

→ Choose proper metrics + reevaluate performance
of classifier

→ Artificially sample a small hold out test set from your data that
contains more of a balance of classes

→ Collect ↑ data

→ Resample data set

Sample w/ replacement

Remove from over-represented data

→ Algorithmic approaches (forest, trees) that tolerate imbalance

When ANALYSING, rely on:

- ROC curves → if balanced

- precision recall → if unbalanced

How much data is needed?

- ↑ n° training data at logarithmic rhythm and fee
 - models like Regression it grows and then reaches a PLATEAU
- "1 in 10" → 10 examples of each level class

→ for NEURAL Networks → ~ 1000 examples

factors that impact on the amount of data needed

- N° of features in dataset relative to i° of uncorrelated / weakly correlated attributes of dataset

Retrospective data + "Shelf life"

⇒ Adding ↑ info ≠ ↑ data

Quantity vs quality

You can overcome data label noise with ↑ volume

10% Noise → 50% data
15% " → double "

⇒ Weak supervision - relative strength of the label → incomplete / inaccurate

→ the test set should have ↓ NOISE as possible
↓
representing the gold standard